Unsupervised Discovery of Semantic Directions in Diffusion Models via h-Space Factorization with Auxiliary Networks

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***Abstract*—Diffusion models have demonstrated remarkable capabilities in high-fidelity image generation. However, understanding and controlling their learned representations, particularly for semantic image editing, remains a challenging task, often requiring supervised data or pre-trained classifiers. This paper presents an unsupervised method for discovering semantic directions within the latent space of pre-trained diffusion models. Our approach focuses on factorizing the bottleneck feature space (h-space) of the UNet architecture. We introduce an auxiliary network that learns to apply directional shifts to these bottleneck features during a specific phase of the denoising process. A regressor network is co-trained to predict the applied direction and magnitude of change from pairs of original and modified images. This, coupled with perceptual and diversity losses, encourages the auxiliary network to discover disentangled and semantically meaningful transformations without explicit attribute labels. We demonstrate our method on the CelebA-HQ dataset using a pre-trained DDPM, successfully identifying interpretable directions corresponding to various facial attributes.**

***Index Terms*—Diffusion Models, Unsupervised Learning, Semantic Directions, Latent Space Manipulation, h-Space Factorization, Auxiliary Networks, Generative Models, Image Editing.**

I. Introduction

DIFFUSION probabilistic models [1], [2] have emerged as powerful generative tools, synthesizing realistic images. They operate via a forward process that gradually adds noise to data, and a reverse process that learns to denoise, generating data from pure noise. While excelling at generation, controllable and semantically meaningful manipulation of their output is an active research area. Existing techniques rely on labeled datasets, pre-trained classifiers, or textual guidance [3], [4], which can be data-intensive or may not capture all desired nuanced variations.

Discovering interpretable directions in the latent spaces of generative models has been explored in Generative Adversarial Networks (GANs) [5], [6]. These directions allow for control over specific image attributes. For diffusion models, the latent representations are distributed across multiple timesteps and within the complex architecture of the denoising network. The intermediate feature maps, particularly at the bottleneck of the UNet (referred to as h-space [7]), are hypothesized to contain rich semantic information.

This work proposes an unsupervised approach to identify and leverage semantic directions within the h-space of a pre-trained diffusion model. Our core idea is to introduce an *Auxiliary Network* that learns to perturb the bottleneck features in specific, consistent ways corresponding to a set of learned directions. A *Regressor Network* is simultaneously trained to recognize these directional modifications by observing pairs of original and edited images. The training regimen is designed such that the auxiliary network is incentivized to produce edits that are (a) classifiable by the regressor, (b) perceptually significant, and (c) diverse across different directions.

Our main contributions include a novel framework for unsupervised discovery of semantic directions in pre-trained diffusion models, achieved by factorizing the UNet's bottleneck h-space. This framework incorporates the design of an auxiliary network that applies learned directional shifts to bottleneck features, integrated into a modified UNet architecture. We further propose a joint training strategy for this auxiliary network and a regressor, leveraging a comprehensive combination of classification, magnitude regression, perceptual, and diversity losses. Finally, we demonstrate our method's ability to find semantically meaningful and disentangled directions on the CelebA-HQ dataset, utilizing a DDPM backbone.

II. Related Work

Denoising Diffusion Probabilistic Models (DDPMs) [2] and related score-based models [8] have set new benchmarks in image generation. Subsequent works have improved sampling speed [9], and explored conditional generation [10], [4]. Understanding the latent representations within these models is crucial for controllability.

A significant body of work exists on finding semantic directions in GAN latent spaces. Methods like InterFaceGAN [6] use attribute classifiers to find linear directions. Unsupervised methods for GANs, such as GANSpace [11], apply PCA to activation layers. Our work seeks similar unsupervised discovery but within the more complex framework of diffusion models.

Editing in diffusion models has been approached through text-guidance [3], [12], classifier guidance [13], or by fine-tuning models on specific tasks. Some works explore editing by manipulating cross-attention maps [14] or by inverting real images into the diffusion latent space [15]. Unsupervised discovery of editing directions is less explored.

The goal of disentangled representation learning is to find latent variables that correspond to distinct, interpretable factors of variation in the data [16]. While many methods focus on VAEs or GANs, the principles of encouraging independence and interpretability are relevant to our goal of finding distinct semantic directions.

Asyrp [7] utilizes the h-space at the bottleneck of diffusion models to discover semantic directions. It employs a CLIP [17] directionality loss to identify these directions. Alternatively, it is also possible to uncover semantic directions using PCA [18].

The idea of using auxiliary networks or manipulating bottleneck features is not new. In GANs, this has been used for style transfer or fine-grained control [19]. Our approach adapts this concept to the UNet architecture of diffusion models, specifically targeting the h-space for semantic factorization. The method presented in this paper is inspired by prior works that factorize GAN latents, but tailored to the unique properties of diffusion models. The unsupervised discovery mechanism through a regressor network shares conceptual similarities with adversarial training or cycle-consistency objectives in other domains.

III. Methodology

Our goal is to discover N_D semantic directions in an unsupervised manner within a pre-trained diffusion model. The core idea is to train an auxiliary network to modify the bottleneck features (h-space) of the UNet denoiser. These modifications are guided by a direction index and a magnitude. A regressor network is simultaneously trained to predict the direction index and magnitude from pairs of original and edited images generated by the diffusion model.

A. System Overview

The system comprises four main components:

1) *Pre-trained Diffusion Model*. We use a standard UNet-based denoiser from a pre-trained diffusion model (e.g., DDPM). Its weights are kept frozen, except where specifically noted for integration.
2) *Auxiliary Network ($\mathcal{A}$)*. A neural network that takes a direction index $d \in \{0, \ldots, N_D - 1\}$ and a scalar magnitude m as input. It outputs a shift tensor Δh which is added to the UNet's bottleneck features.
3) *Modified UNet (UNet_{mod})*. A wrapper around the pre-trained UNet that incorporates the auxiliary network. During specific timesteps of the reverse diffusion process, it applies the shift Δh from $\mathcal{A}$ to the bottleneck features.
4) *Regressor Network ($\mathcal{R}$)*. A classifier and regressor that takes a pair of images (original I_o, edited I_e) and predicts the direction index $\hat{d}$ and magnitude $\hat{m}$ of the edit.

The overall training process involves generating edited images using UNet_{mod} and then training both $\mathcal{A}$ and $\mathcal{R}$ based on the regressor's predictions and other losses.

B. Auxiliary Network

The Auxiliary Network, denoted as $\mathcal{A}$, is tasked with producing a modification, or shift tensor Δh, specifically tailored to alter the UNet's bottleneck features. This shift is intended to correspond to a learned semantic direction(see Figure 2). In our setup, $\mathcal{A}$ is realized as a multi-layer fully connected network.

Its inputs are twofold: a batch of direction vectors $\mathbf{v} \in R^{B \times N_D}$ (where B is the batch size and N_D is the number of distinct semantic directions, with each vector typically being a one-hot encoding selecting a single direction), and a scalar magnitude m that dictates the strength of the applied edit.

Internally, the network processes the direction vector $\mathbf{v}$ through a sequence of fully connected (FC) layers, each followed by Batch Normalization (BN) and an Exponential Linear Unit (ELU) activation function. This can be represented as:

$$\text{features}_{\text{intermediate}} = \ldots \text{ELU}(\text{BN}(\text{FC}_1(\mathbf{v}))) \ldots \quad (1)$$

$$\text{shift}_{\text{flat}} = \text{FC}_{\text{out}}(\text{features}_{\text{intermediate}}) \quad (2)$$

The initial output, $\text{shift}_{\text{flat}} \in R^{L \times D}$, where L is the number of layers the shift is applied and D is the flattened dimension of the bottleneck. This vector is then reshaped to match the three-dimensional structure (channels C_b, height H_b, width W_b) of the UNet's bottleneck feature maps. Finally, this reshaped tensor is scaled by the input magnitude m to produce the actual shift Δh:

$$\Delta h = \text{reshape}(\text{shift}_{\text{flat}}) \times m \quad (3)$$

This Δh is then added to the original bottleneck features during the diffusion model's denoising process to effect the semantic change. Note that Δh contains different shift directions for each edited layer. This mechanism is essential to imitate conditioning the auxiliary network on the time step.

C. Modified UNet and Denoising Process

To incorporate semantic edits, we adapt the pre-trained UNet by creating a 'ModifiedUNet'. This adaptation involves intervening in the UNet's forward pass specifically at its 'mid_block', which outputs the bottleneck features, denoted as h. When an edit is desired and the current denoising timestep t is within a designated window (from $T_{\text{target}} - \Delta T$ up to T_{target}), the shift Δh's corresponding shift vector Δh_i is added to these features:

$$h_{\text{mod}} = h + \Delta h_l, \; l \in [0, L-1] \quad (4)$$

These modified bottleneck features, h_{mod}, then proceed through the remaining layers of the UNet.

The iterative denoising process, which generates an image from noise, is managed by a 'CustomPretrainedDiffusionModel' using a standard scheduler like DDIM [9]. A specific update rule is employed when an edit is active within the designated timesteps. For a given latent x_t at timestep t:

1) The standard, frozen UNet provides an original noise prediction, $\epsilon_\theta(x_t, t)$.
2) The 'ModifiedUNet', utilizing the auxiliary network $\mathcal{A}$ to alter h, produces an edited noise prediction, $\epsilon'_\theta(x_t, t)$.
3) An estimate of the clean image, x_0^{edited}, is derived using this edited noise: $x_0^{\text{edited}} = (x_t - \sqrt{1-\bar{\alpha}_t}\epsilon'_\theta(x_t, t))/\sqrt{\bar{\alpha}_t}$, where $\bar{\alpha}_t$ is the cumulative noise schedule product.
4) The subsequent latent, x_{t-1}, is then formed by combining this edited clean image estimate with the original noise prediction:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} x_0^{\text{edited}} + \sqrt{1-\bar{\alpha}_{t-1}}\epsilon_\theta(x_t, t) \quad (5)$$

This approach aims to infuse the semantic directionality from ϵ'_θ into the predicted x_0, while retaining the overall structural integrity guided by ϵ_θ for the noise component of the next step.

If no edit is intended, or if the current timestep t is outside the specified modification window, the denoising proceeds conventionally using only the original noise prediction $\epsilon_\theta(x_t, t)$ to compute x_{t-1}.

D. Regressor Network

The Regressor Network, denoted as $\mathcal{R}$, is designed to process a pair of images (I_o, I_e), where I_o is generated without auxiliary network intervention and I_e is generated with a specific direction d and magnitude m applied via $\mathcal{A}$. Built upon a ResNet-18 architecture, $\mathcal{R}$ features a modified first convolutional layer to accommodate $2 \times C_{img}$ channels (e.g., 6 for RGB images), with features from the average pooling layer subsequently fed into two distinct linear heads: a Type Estimator responsible for predicting logits $\hat{l}$ for the N_D directions, and a Shift Estimator that predicts a scalar magnitude $\hat{m}$.

E. Training and Loss Functions

The training proceeds in steps. In each step:

1) A batch of initial noise latents z_0 is sampled.
2) For each z_{0i} in the batch, M random directions d_{ij} and M random magnitudes m_{ij} (within a predefined range, with random signs) are sampled.
3) For each (z_{0i}, d_{ij}, m_{ij}), an edited image I_e^{ij} is generated using UNet_{mod}.
4) An original image I_o^i is generated from z_{0i} without auxiliary network intervention. These are repeated M times to pair with the M edits per initial noise.
5) The regressor $\mathcal{R}$ processes pairs (I_o^i, I_e^{ij}) to get predictions $\hat{l}_{ij}, \hat{m}_{ij}$.

The combined loss function $\mathcal{L}_{\text{total}}$ is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{shift}} + w_{\text{lpips}}\mathcal{L}_{\text{LPIPS}} + w_{\text{div}}\mathcal{L}_{\text{div}} \quad (6)$$

Here, $\mathcal{L}_{\text{cls}} = \text{CrossEntropyLoss}(\hat{\ell}, d_{\text{true}})$ encourages the regressor to correctly classify which direction was applied (and in turn pushes $\mathcal{A}$ to make its edits distinguishable); $\mathcal{L}_{\text{shift}} = \text{L1Loss}(\hat{m}, m_{\text{true}})$ encourages accurate magnitude prediction (and proportional editing by $\mathcal{A}$); $\mathcal{L}_{\text{LPIPS}} = \text{LPIPS}(I_o, I_e)$ is a perceptual loss that ensures edits remain realistic yet perceptually meaningful; and finally

$$\mathcal{L}_{\text{div}} = -\frac{1}{B\binom{M}{2}} \sum_{i=1}^{B} \sum_{j_1<j_2} \text{mean}\big((I_e^{ij_1} - I_e^{ij_2})^2\big)$$

penalizes similarity between different edited outputs from the same base noise (promoting diversity and disentanglement among the learned directions). Gradients of $\mathcal{L}_{\text{total}}$ flow through both the Auxiliary Network $\mathcal{A}$ and the Regressor $\mathcal{R}$.

The gradients from $\mathcal{L}_{\text{total}}$ update the parameters of both the Auxiliary Network $\mathcal{A}$ and the Regressor Network $\mathcal{R}$.

Even though incorporating a per-direction magnitude is key to discovering meaningful edit axes, using large values too early can destabilize training and hinder the classifier's convergence. To address this, we gradually increase the maximum magnitude according to an exponential schedule:

$$\text{frac_done} = \frac{\text{epoch}}{N_{\text{epochs}} - 1}, \qquad \text{m}_{\text{current}} = m_{\text{init}}\left(\frac{m_{\text{final}}}{m_{\text{init}}}\right)^{\text{frac_done}}.$$

This allows the model to first master small, high-fidelity edits and then progressively handle larger, more dramatic transformations.

IV. Experiments

A. Method and Dataset Description

The core of our experimental setup is the 'DiffusionModel' class, which orchestrates the interaction between the 'CustomPretrainedDiffusionModel' (containing the 'ModifiedUNet' and 'AuxiliaryNetwork') and the 'ResnetRegressor'. The training loop ('train_step') generates batches of original and edited images by applying M different randomly chosen directions and magnitudes to each of B initial noise samples. These image pairs are then used to train the regressor and, through backpropagation, the auxiliary network. We use the CelebA-HQ dataset [20], which consists of 30,000 high-quality images of celebrity faces. The images are pre-processed to a resolution of 256×256 pixels. The pre-trained diffusion model backbone is 'google/ddpm-ema-celebahq-256' [2], which was trained on this dataset.

B. Details of Trained Models and Configuration

The diffusion model backbone is 'google/ddpm-ema-celebahq-256', a UNet with 512 channels and 8×8 spatial features at its bottleneck. Checkpointing is used for efficient training. We utilize 50 of the 1000 denoising steps, with edits (Δh) applied between steps 800 and 700. The 'AuxiliaryNetwork' takes a one-hot encoded 'num_directions' as input.

It comprises four fully connected layers (‘inner_dim=1024‘) with ‘BatchNorm1d‘, ‘ELU‘ activation, and skip connections (output of layer k added to input of $k+1$), outputting a $(512, 8, 8)$ shift tensor for the diffusion model's bottleneck. The ‘ResnetRegressor‘ uses a ‘resnet18‘ base. Its ‘conv1‘ layer is modified for 6 input channels (concatenated original and edited $256 \times 256 \times 3$ images). Two linear heads on the ‘avgpool‘'s 512-dimensional feature vector predict ‘num_directions‘ logits and a single shift magnitude. Detailed training hyperparameters are in Appendix A. Training was performed on a single NVIDIA H100 GPU. PyTorch [21] and the Diffusers library [22] were used for implementation.

V. Results

A. Qualitative and Quantitative Evaluation

1) Training Losses: During training, we monitored the constituent loss terms. Typical values observed after convergence were:

- $\mathcal{L}_{\text{cls}}$ (Cross-Entropy Loss for direction classification): Converged to low values (e.g., 0.1-0.5), indicating the regressor could distinguish the directions applied by the auxiliary network.
- $\mathcal{L}_{\text{shift}}$ (L1 Loss for magnitude): Also converged (e.g., 0.2-0.6), showing the regressor learned to estimate the strength of edits.
- $\mathcal{L}_{\text{LPIPS}}$ (Perceptual Loss): Maintained a reasonable value (e.g., 0.05-0.15 after weighting), ensuring edits were perceptually aligned with the original images while still being distinct.
- $\mathcal{L}_{\text{div}}$ (Diversity Loss): Decreased (as it's negatively weighted), indicating that edits for different directions from the same noise became more distinct.

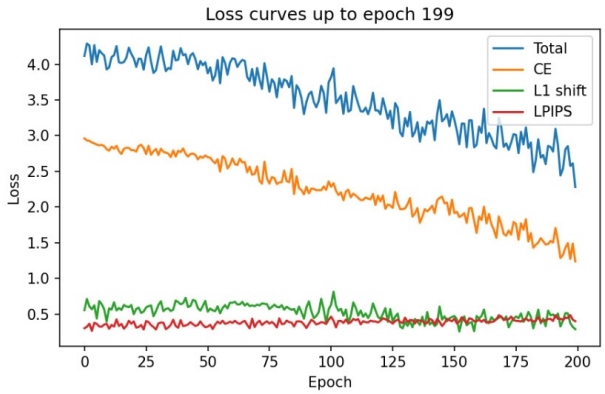


Fig. 1: Training Loss

Figure 1 shows the training loss change. The overall loss $\mathcal{L}_{\text{total}}$ showed a steady decrease over epochs, for instance, from initial high values to values around 1.5-3.0 after significant training. An example log from a late epoch: ‘[Epoch 190] total_loss=2.15, cls_loss=0.35, shift_loss=0.40, lpips_loss_val=0.10 (weighted: 0.20), diversity_loss_val=-2.40 (weighted: -1.20)‘. Note that ‘lpips_loss_val‘ and ‘diversity_loss_val‘ are shown before their respective weighting factors w_{lpips} and w_{div} in this example.

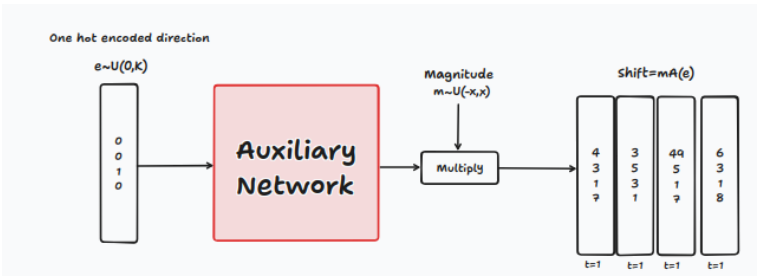
2) Visual Inspection of Discovered Semantic Directions: The primary evaluation of our unsupervised method is qualitative, by inspecting the visual changes corresponding to the discovered directions when applied to random initial noise latents. The following figures illustrate examples of such interpolations for several discovered directions. Each sequence shows the effect of varying the magnitude of the edit along the respective direction.

Visual inspection reveals that the discovered directions often correspond to interpretable semantic attributes. For instance, directions were found to affect Age (Figures 3 and 4), perceived Gender (Figure 5), Hair Length/Style (Figure 6), and features like Jaw Line, sometimes entangled with age (Figure 7). Other commonly found directions include smile, pose, eye gaze, and lighting. While exact semantics can vary between unsupervised training runs, consistently interpretable directions are typically discovered.

Qualitative results demonstrate our unsupervised method successfully discovers multiple, seemingly disentangled semantic directions in a pre-trained DDPM's h-space. The combination of the regressor (identifiability), LPIPS loss (perceptual realism), and diversity loss (distinctness among directions) appears highly effective. A primary strength is its unsupervised nature, requiring no attribute labels or pre-trained classifiers. Furthermore, the method is applicable to existing pre-trained diffusion models, as it only necessitates training relatively smaller auxiliary and regressor networks. Finally, the diversity loss and distinct directional signals notably promote disentanglement, ensuring edits along one direction minimally affect attributes associated with others.

Despite its strengths, interpretability remains a problem, as assigning labels to the discovered semantic directions is still a human interpretation step, and their exact nature can vary across training runs. Moreover, extreme magnitudes can sometimes lead to artifacts in the quality of edits. Furthermore, developing robust quantitative metrics for the semantic correctness or disentanglement of unsupervisedly discovered directions, without relying on external classifiers, remains a challenge. Additionally, computational limitations restricted us to 16 directions, which is an issue since many feature directions exist, and using only a few may result in discovering directions that edit multiple features simultaneously. Future exploration could involve more structured modifications of the h-space, beyond modifying the entire bottleneck, potentially yielding different types of control.

3) Quantitative Study: For quantitative study, Reconstructor Classifier Accuracy (RCA) [19] is used where the data is generated for each edit directions which are treated as classes. Then with the given data ResNet-18 with edited last layer is trained on it and accuracy is given. Accuracy of this result for 16 directions, 200 samples per each



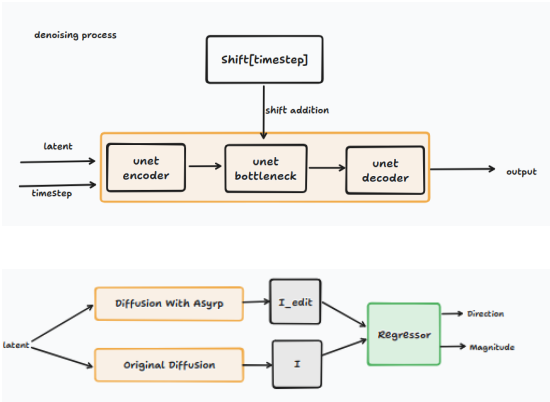


Fig. 2: *Left.* The Auxiliary Network takes a sampled one-hot direction $e \sim U(0, K)$ and produces a set of per-time-step shifts of shape (C, H, W). We sample a scalar magnitude $m \sim U(-x, x)$ and multiply it against each shift, yielding time-dependent bottleneck adjustments. Multiple shifts are output as a batch to encode the denoise timestep prior. *Top right.* During denoising at time-step t, the scaled shift is injected into the UNet bottleneck via the edited UNet block (UNet_edit), and the Asyrp [7] operator is applied to guide the image update. *Bottom right.* Both the edited (I_{edit}) and original (I) diffusion outputs are fed into a ResNet based regressor network, which predicts the semantic direction index and the interpolation magnitude.

Fig. 3: Interpolation along a discovered direction primarily affecting perceived age in a female subject. From left to right, the magnitude of the "aging" edit increases.

Fig. 4: Interpolation along a discovered direction primarily affecting perceived age in a male subject, also showing changes in hair color. From left to right, the magnitude of the "aging" edit increases.

Fig. 5: Interpolation along a discovered direction affecting perceived gender. The transition from a more feminine appearance (left) to a more masculine appearance (right) is observed.

direction for testing and 600 for training. 68.15% accuracy is achieved. Where random gives 6% accuracy. Hence, we showed that the learned directions are interpretable by the resnet.

4) Human Oracle Study on Direction Interpretability: To further assess the interpretability and consistency of the discovered semantic directions, a small-scale survey was

Fig. 6: Interpolation along a discovered direction affecting hair length and style in a male subject. From left to right, the hair appears to shorten and become neater.

Fig. 7: Interpolation along a discovered direction intended to modify the jawline in a female subject.

conducted with 13 human oracles. Oracles were shown the interpolation sequences (similar to Figures 3-7) for five distinct discovered directions and asked to identify the primary semantic change from a predefined list: Age, Gender, Hair Length, or Jaw Line. The results are summarized below.

a) Direction 1: Age (Female Subject): The interpolation shown in Figure 3 was evaluated. The true underlying semantic change is Age. Oracle responses are detailed in Table I.

b) Direction 2: Age (Male Subject): The interpolation shown in Figure 4 was evaluated. The true underlying semantic change is Age. Oracle responses are detailed in Table II.

c) Direction 3: Gender: The interpolation shown in Figure 5 was evaluated. The true underlying semantic change is Gender. Oracle responses are detailed in Table III.

TABLE I: Human Oracle Evaluation for Direction 1 (True: Age)

Oracle ID	Perceived Direction	Correct?
1	Age	Yes
2	Age	Yes
3	Age	Yes
4	Age	Yes
5	Jaw Line	No
6	Age	Yes
7	Age	Yes
8	Age	Yes
9	Age	Yes
10	Age	Yes
11	Age	Yes
12	Age	Yes
13	Hair Length	No (Mix-up)

TABLE II: Human Oracle Evaluation for Direction 2 (True: Age)

Oracle ID	Perceived Direction	Correct?
1	Age	Yes
2	Age	Yes
3	Age	Yes
4	Age	Yes
5	Age	Yes
6	Age	Yes
7	Age	Yes
8	Age	Yes
9	Hair Length	No
10	Age	Yes
11	Age	Yes
12	Age	Yes
13	Hair Length	No (Mix-up)

TABLE III: Human Oracle Evaluation for Direction 3 (True: Gender)

Oracle ID	Perceived Direction	Correct?
1	Gender	Yes
2	Gender	Yes
3	Gender	Yes
4	Gender	Yes
5	Gender	Yes
6	Gender	Yes
7	Gender	Yes
8	Gender	Yes
9	Gender	Yes
10	Gender	Yes
11	Gender	Yes
12	Gender	Yes
13	Gender	Yes

d) Direction 4: Hair Length (Male Subject): The interpolation shown in Figure 6 was evaluated. The true underlying semantic change is Hair Length. Oracle responses are detailed in Table IV.

e) Direction 5: Jaw Line (Female Subject): The interpolation shown in Figure 7 was evaluated. The intended underlying semantic change for this discovered direction vector was Jaw Line, though visually it is similar to Figure 3. Oracle responses are detailed in Table V.

TABLE IV: Human Oracle Evaluation for Direction 4 (True: Hair Length)

Oracle ID	Perceived Direction	Correct?
1	Hair Length	Yes
2	Hair Length	Yes
3	Hair Length	Yes
4	Hair Length	Yes
5	Hair Length	Yes
6	Hair Length	Yes
7	Age	No
8	Hair Length	Yes
9	Hair Length	Yes
10	Hair Length	Yes
11	Hair Length	Yes
12	Hair Length	Yes
13	Age	No (Mix-up)

TABLE V: Human Oracle Evaluation for Direction 5 (Intended: Jaw Line)

Oracle ID	Perceived Direction	Correct (vs Intended)?
1	Age	No
2	Age	No
3	Jaw Line	Yes
4	Age	No
5	Age	No
6	Age	No
7	Jaw Line	Yes
8	Age	No
9	Age	No
10	Age	No
11	Age	No
12	Jaw Line	Yes
13	Jaw Line	Yes

f) Summary of Human Oracle Study: Human oracles showed high agreement on perceived semantic changes. For distinct visual changes like Gender (Table III), Age (Tables I, II), and Hair Length (Table IV), oracle accuracy was high. Direction 5 (Jaw Line, Table V), visually similar to an aging effect (Figure 3), was mostly perceived as 'Age', indicating potential entanglement, though some oracles identified 'Jaw Line'. Oracle 13, designed for an error condition, consistently confused 'Age' and 'Hair Length' as intended but correctly identified other attributes. Overall, the study supports that the model discovers largely human-interpretable directions, though some may be less disentangled or exhibit coupled changes.

VI. Conclusion

We presented an unsupervised method to discover semantic editing directions in pre-trained diffusion models. By perturbing bottleneck features using an auxiliary network and having a regressor identify these changes, our system learns disentangled, semantically meaningful directions without explicit supervision. Experiments on CelebA-HQ demonstrated learning of attributes like smile, age, and pose. This work enhances model interpretability and controllability, facilitating intuitive, label-free image editing. Future research could explore scaling to more directions, refining semantic disentanglement evaluation, and investigating alternative factorization methods.

APPENDIX A
TRAINING HYPERPARAMETERS

TABLE VI: Training Hyperparameters

Hyperparameter	Value
Directions (N_D)	16
Target Timestep (T_{target})	800 (of 1000 DDPM steps)
Edit Duration (ΔT)	100 timesteps (800 to 701)
Inference Steps (DDIM)	100
Learning Rate (Adam)	1×10^{-4}
Batch Size (B)	2
Directions per Noise (M)	16
Max Magnitude	2.2
LPIPS Weight (w_{LPIPS})	2.0
Diversity Weight (w_{div})	0.5
Epochs	200
Number of Denoising Steps (L)	50

REFERENCES

[1] J. Sohl-Dickstein, E. A. Weiss, N. Maheswaranathan, and S. Ganguli, "Deep unsupervised learning using nonequilibrium thermodynamics," 2015. [Online]. Available: https://arxiv.org/abs/1503.03585

[2] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," 2020. [Online]. Available: https://arxiv.org/abs/2006.11239

[3] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen, "Hierarchical text-conditional image generation with clip latents," 2022. [Online]. Available: https://arxiv.org/abs/2204.06125

[4] A. Nichol, P. Dhariwal, A. Ramesh, P. Shyam, P. Mishkin, B. McGrew, I. Sutskever, and M. Chen, "Glide: Towards photorealistic image generation and editing with text-guided diffusion models," 2022. [Online]. Available: https://arxiv.org/abs/2112.10741

[5] A. Radford, L. Metz, and S. Chintala, "Unsupervised representation learning with deep convolutional generative adversarial networks," 2016. [Online]. Available: https://arxiv.org/abs/1511.06434

[6] Y. Shen, J. Gu, X. Tang, and B. Zhou, "Interpreting the latent space of gans for semantic face editing," 2020. [Online]. Available: https://arxiv.org/abs/1907.10786

[7] M. Kwon, J. Jeong, and Y. Uh, "Diffusion models already have a semantic latent space," 2023. [Online]. Available: https://arxiv.org/abs/2210.10960

[8] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," 2021. [Online]. Available: https://arxiv.org/abs/2011.13456

[9] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," 2022. [Online]. Available: https://arxiv.org/abs/2010.02502

[10] P. Dhariwal and A. Nichol, "Diffusion models beat gans on image synthesis," 2021. [Online]. Available: https://arxiv.org/abs/2105.05233

[11] E. Härkönen, A. Hertzmann, J. Lehtinen, and S. Paris, "Ganspace: Discovering interpretable gan controls," 2020. [Online]. Available: https://arxiv.org/abs/2004.02546

[12] C. Meng, Y. Song, J. Song, J. Wu, J.-Y. Zhu, and S. Ermon, "SDEdit: Image synthesis and editing with stochastic differential equations," *International Conference on Learning Representations*, 2021.

[13] P. Dhariwal and A. Nichol, "Diffusion models beat gans on image synthesis," 2021. [Online]. Available: https://arxiv.org/abs/2105.05233

[14] A. Hertz, R. Mokady, J. Tenenbaum, K. Aberman, Y. Pritch, and D. Cohen-Or, "Prompt-to-prompt image editing with cross attention control," 2022. [Online]. Available: https://arxiv.org/abs/2208.01626

[15] J. Choi, S. Kim, Y. Jeong, Y. Gwon, and S. Yoon, "Ilvr: Conditioning method for denoising diffusion probabilistic models," 2021. [Online]. Available: https://arxiv.org/abs/2108.02938

[16] Y. Bengio, A. Courville, and P. Vincent, "Representation learning: A review and new perspectives," 2014. [Online]. Available: https://arxiv.org/abs/1206.5538

[17] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," 2021. [Online]. Available: https://arxiv.org/abs/2103.00020

[18] R. Haas, I. Huberman-Spiegelglas, R. Mulayoff, S. Graßhof, S. S. Brandt, and T. Michaeli, "Discovering interpretable directions in the semantic latent space of diffusion models," 2024. [Online]. Available: https://arxiv.org/abs/2303.11073

[19] A. Voynov and A. Babenko, "Unsupervised discovery of interpretable directions in the gan latent space," 2020. [Online]. Available: https://arxiv.org/abs/2002.03754

[20] T. Karras, T. Aila, S. Laine, and J. Lehtinen, "Progressive growing of gans for improved quality, stability, and variation," *CoRR*, vol. abs/1710.10196, 2017. [Online]. Available: http://arxiv.org/abs/1710.10196

[21] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, "Pytorch: An imperative style, high-performance deep learning library," 2019. [Online]. Available: https://arxiv.org/abs/1912.01703

[22] P. von Platen, S. Patil, A. Lozhkov, P. Cuenca, N. Lambert, K. Rasul, M. Davaadorj, D. Nair, S. Paul, W. Berman, Y. Xu, S. Liu, and T. Wolf, "Diffusers: State-of-the-art diffusion models," https://github.com/huggingface/diffusers, 2022.